dissolves peptidoglycan. However, Gram negative bacteria are refractory to this enzyme because large protein molecules of enzyme cannot penetrate the outer membrane. Only when outer membrane is damaged the enzyme can penetrate.

- Porins allow the smaller molecules, such as amino acids, monosaccharides to pass across.
- Adhesions are export sites for newly synthesised LPS and porins, and are sites at which pili and flagella are made.

7.3.3 Structure of Cytoplasmic Membrane

Immediately beneath the cell wall is the cytoplasmic membrane also known as plasma membrane or cell membrane. It is composed of phospholipids and proteins. The phospholipids form a bilayer. Integral proteins are embedded within this bilayer. Surface proteins or peripheral proteins are loosely attached to the bilayer. The lipid matrix of the membrane has fluidity, allowing the components to move around laterally. In eubacteria, the phospholipids are phosphoglycerides, in which straight chain fatty acids are ester linked to glycerol. In archaeobacteria, the lipids are polyisoprenoid branched-chain lipids, in which long-chain branched alcohols (phytanols) are ether linked to glycerol.

Functions of the cell membrane

- Prokaryotes do not have intracellular membrane bound organelles as present in eukaryotic organelles. Thus cell membrane provides a site for functions such as energy reactions, nutrient processing and synthesis.
- It regulates transport, the passage of nutrients into the cell and the discharge of wastes. It is a selectively permeable membrane.
- It is also involved in secretion or discharge of a metabolic product into extracellular environment.
- Cell membrane is an important site for a number of metabolic activities. Most enzymes of respiration and ATP synthesis reside in the cell membrane since prokaryotes lack mitochondria.

Significance of cell envelope

- It has toxic properties (Example: LPS)
- It stimulates antibody production by immune system
- The cell walls of many pathogens have components that contribute to their pathogenicity. Example mycolic acids of *Mycobacterium tuberculosis*
- Cell wall is a site of action of several antibiotics.
- Many of the serological properties of Gram negative bacteria are attributable to O antigens; they can also serve as receptors for bacteriophage attachment.

7.4 Structures Internal to Cell Membrane of Bacteria

Cytoplasm is called as the internal matrix of the cell inside the cell membrane. Its major component is water (70-80%). It also contains proteins carbohydrates, lipids, inorganic ions, and certain low molecular weight compounds. Inorganic ions are present in much higher concentrations in cytoplasm than in most media.

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